

AN INTERNSHIP REPORT
OF
C++ PROGRAMMING INTERNSHIP

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.



In partial fulfilment of the requirement of degree of
B-Tech Computer Science Engineering

Submitted By
SHOORVEER PRATAP AGRAHARI
Enrollment No.: 25SCS1003004868
Semester III , Batch: 2025-29

Organization: CodeAlpha

Internship Duration: 10 August 2026 – 10 September 2026

CANDIDATE'S DECLARATION

I, SHOORVEER PRATAP AGRAHARI, hereby declare that the internship report titled “C++ Programming Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

I further declare that the work presented in this report is based on my learning and practical work completed during the CodeAlpha Virtual Internship Program in C++ Programming. The projects described in this report, including the CGPA Calculator and Login & Registration System, were developed as internship tasks using C++ programming concepts.

All references used for preparing this report have been acknowledged. I affirm that this report has not been submitted to any other institute for the same degree/diploma requirement, and I shall be responsible for any copyright violation arising from unauthorized use of material.

Student Name: SHOORVEER PRATAP AGRAHARI

Enrollment No.: 25SCS1003004868

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the internship program. I am thankful for the guidance, encouragement, constructive feedback and learning opportunities that helped me complete the internship successfully.

I sincerely thank CodeAlpha for providing me the opportunity to participate in the Virtual Internship Program in C++ Programming. The internship provided a practical environment to apply programming concepts through task based projects and helped me improve my understanding of C++ programming, problem solving, input validation, file handling and menu driven applications.

I am also thankful to my University, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement throughout my academic and internship work.

Finally, I express my gratitude to my parents and everyone who supported me during the completion of this internship.

Student Name: SHOORVEER PRATAP AGRAHARI

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3. INTERNSHIP COMPLETION CERTIFICATE



The certificate confirms successful completion of the CodeAlpha Virtual Internship Program in C++ Programming from 10th August 2026 to 10th September 2026. The certificate is issued in the name of Shoorveer Pratap Agrahari and is dated 10th September 2026.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the one-month CodeAlpha Internship Program undertaken in the domain of C++ Programming. The internship was carried out from 10th August 2026 to 10th September 2026 and was completed as part of the academic and practical learning requirements of the Bachelor of Technology in Computer Science and Engineering.

The main purpose of the internship was to strengthen programming fundamentals through practical, task-based development. During the internship, C++ was used to create console applications that demonstrate input handling, calculations, conditional statements, loops, functions, file handling, validation and menu-driven programming.

Two completed projects are documented in this report: a CGPA Calculator and a Login & Registration System. These projects provided practical exposure to converting requirements into working programs, testing different input cases and maintaining readable project documentation.

4.2 Organization / Program Profile

CodeAlpha provided the Virtual Internship Program in C++ Programming under which the internship tasks were completed. The program offered a structured, practical learning environment in which programming concepts were applied through project-based tasks. The completion certificate issued by CodeAlpha confirms successful completion of the program.

4.3 Problem Statement

The internship focused on solving practical programming problems through simple and reliable console-based applications. The first problem involved calculating a student's CGPA from subject-wise grade points and credit hours. The second involved implementing a basic registration and login workflow with username validation, duplicate-user checking and local file handling.

4.4 Project Objectives

- To strengthen practical knowledge of C++ programming.
- To implement console-based applications from defined requirements.
- To practice variables, data types, operators, conditions, loops and functions.
- To implement input validation and handle incorrect user input.
- To understand basic file handling through a login and registration project.
- To test, document and present completed programming tasks professionally.

4.5 Scope of the Project

The scope of the internship work covered the design and implementation of small C++ console applications. The CGPA Calculator covered subject wise data entry, weighted grade point calculation, credit-hour calculation, validation and final CGPA presentation. The Login & Registration System covered registration, duplicate username detection, login verification, invalid credential handling and local file-based credential storage.

The projects were developed for educational and internship purposes and are intended to demonstrate programming concepts rather than production-grade software deployment.

4.6 Technologies and Tools Used

- Programming Language: C++
- Compiler: GCC / MinGW
- IDE: Visual Studio Code
- Operating Platform: Windows
- Version Control: Git
- Repository Hosting: GitHub
- File Handling: C++ file streams for local data storage in the Login & Registration task

4.7 System Architecture

The projects follow a simple console-based application architecture. The user interacts with a menu or input prompts, the program validates and processes the supplied data, and the result is displayed through the console. The Login & Registration System additionally uses a local text file for storing and retrieving registered credentials.

- Input Layer: Accepts user details, subject information, grades, credits or login credentials.
- Validation Layer: Checks input values, invalid credentials and duplicate usernames where applicable.
- Processing Layer: Performs CGPA calculations or authentication logic.
- Storage Layer: Uses local file handling for the Login & Registration System.
- Output Layer: Displays calculated results, status messages and validation feedback.

Task 1 — CGPA Calculator: The core calculation uses the weighted average formula: $CGPA = \Sigma(\text{Credit Hours} \times \text{Grade Points}) / \Sigma(\text{Credit Hours})$.

Task 2 — Login & Registration: The system provides Register, Login and Exit options. Registration checks for an existing username before storing new credentials, while Login compares the entered credentials with stored data.

4.8 Methodology

The internship followed a practical, task-oriented methodology. Each project was developed by first understanding the required functionality, then implementing the logic in C++, testing different input cases and documenting the completed work.

- Requirement Understanding: Identifying the expected inputs, processing steps and outputs for each task.
- Design: Planning the menu flow, calculations, validation rules and file-handling approach.
- Implementation: Writing the C++ source code using functions, loops, conditions, strings, arithmetic operations and file handling where required.
- Testing: Running normal and invalid cases, including validation and duplicate-user scenarios.
- Documentation: Preparing README files, screenshots, project structure and GitHub repositories for the completed tasks.
- Version Control: Maintaining the project work using Git and GitHub.

4.9 Expected Outcomes

- Improved understanding of C++ programming fundamentals.
- Practical experience in developing console-based applications.
- Better understanding of input validation and error handling.
- Practical exposure to file handling through a login system.
- Improved debugging, testing and problem-solving skills.
- Experience in documenting projects and managing source code using GitHub.
- Completion of the CodeAlpha C++ Programming Internship with project evidence.

4.10 Certificates of Completion and Project Evidence

The CodeAlpha Internship Completion Certificate confirming successful completion of the C++ Programming internship is attached in Chapter 3 of this report. Project evidence for the completed tasks includes source code files, README documentation and screenshots demonstrating program input, output and validation.

The two documented internship tasks are: (1) CGPA Calculator and (2) Login & Registration System. Their project documentation includes the implemented features, technologies, formulas or workflows, testing details and screenshots.

5. BIBLIOGRAPHY / REFERENCES

1. CodeAlpha, “Virtual Internship Program C++ Programming,” Internship Completion Certificate issued to Shoorveer Pratap Agrahari on 10th September 2026.
2. CodeAlpha C++ Programming Internship Task 1: CGPA Calculator :- project source code, README and testing screenshots.
3. CodeAlpha C++ Programming Internship Task 2: Login & Registration System :- project source code, README and testing screenshots.
4. ISO C++ Foundation, C++ language and standard library reference materials.
5. IILM University, School of Computer Science and Engineering, academic internship/report requirements and guidelines.
6. URL Verication - <https://www.codealpha.tech/>
7. Github Intren Repo. link - https://github.com/shoorveer45/CodeAlpha_CPP_Internship